LEA MARIA WIESNER, Dr.-Ing.

SCIENTIST & PROJECT LEAD (PhD) | LIFE SCIENCES

CONTACT

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SUMMARY

Scientist and project lead with a PhD in Chemical Process Engineering and experience delivering multidisciplinary biomedical and engineering research. Combines hands-on experimental work in microfluidics, biomaterials and tissue engineering with responsibility for projects, budgets, laboratory operations, research quality and stakeholder coordination. Enjoys bringing structure to complex scientific work, improving how teams and experiments are organised, and contributing to research with a clear practical purpose.

SKILLS

Rigorous: Reproducibility | Data quality | Documentation | SOPs | Experimental records | Research data management

Organised: Project coordination | Timeline and budget monitoring | Laboratory operations | Procurement | Prioritisation | Process improvement

Collaborative: Team leadership | Stakeholder coordination | International collaboration | Student supervision | Training and development | Clear communication

Analytical: Experimental design and data analysis | Data comparison | Systematic troubleshooting | Scientific interpretation | Literature review | Evidence-based decision-making

Technical: Microfluidics | Biomaterials | Hydrogels | Tissue engineering | Cell-based systems | Microscopy

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | 03/2025 – 06/2025

UVA – University of Virginia (USA), *Griffin Lab*, Biomedical Engineering

- Conducted independent tissue-engineering research involving biomaterials and cell-based systems within an international, multidisciplinary biomedical engineering group.
- Adapted quickly to the new research environment and established productive collaborations across disciplines.

Project Coordinator & Research Associate | 06/2020 – 02/2025

RWTH Aachen University, *Chair of Chemical Process Engineering*

- Led the scientific and operational delivery of a German Research Foundation-funded biomedical research project, with responsibility for budget, procurement, resources, timelines, documentation and reporting (4 years, €100,000 annual budget). Completed the project on time and within budget.
- Managed up to 3 parallel research projects using structured work packages, Gantt-based planning and progress tracking, adapting priorities to changing results, dependencies and equipment availability.
- Planned, conducted and analysed experimental work in microfluidics, hydrogels and biomaterials. Managed a custom-built microgel fabrication system and investigated whether variation arose from the method, materials or equipment before adapting the process.
- Developed and implemented SOPs, standardised experimental records and structured laboratory workflows that improved reproducibility and helped new students and researchers apply methods consistently. Used a digital research data management platform to organise experiment types, process parameters and contextual metadata.
- Coordinated daily laboratory operations for around 10 active users, including equipment oversight, procurement, suppliers and sample management (5 years).
- Provided scientific coordination for 6 PhD researchers without formal line authority, helping to define objectives, clarify responsibilities, compare evidence and communicate recommendations to the wider project group. Also coordinated a research group of around 20 researchers, introducing structured meetings and documented follow-up that made responsibilities and progress more transparent.
- Led an international industry collaboration in the field of haemodialysis, coordinating responsibilities, timelines and research outputs across academic and industry stakeholders (1.5 years).
- Recruited, onboarded and supervised more than 40 student researchers, allocating work, explaining experimental procedures, reviewing data and supporting students to apply methods independently.
- Published 6 peer-reviewed papers and presented findings to scientific and interdisciplinary audiences. Provided underlying datasets for reported results and documented detailed methods and the source and version of publicly available analysis code.

Student Assistant | 03/2018 – 02/2020
DWI – Leibniz Institute for Interactive Materials

- Performed microfluidic and cell-culture experiments within interdisciplinary research projects, following established protocols and documenting results carefully.
- Supported shared laboratory activities and developed a strong foundation in careful experimental work and reliable laboratory practice.

EDUCATION

PhD (Dr.-Ing.), *summa cum laude* | 06/2020 – 04/2025
Chemical Process Engineering, RWTH Aachen University
Dissertation: *Tailoring the Characteristics of Complex-Shaped Microgels*

Master of Science, *German grade: 1.3 (with distinction)* | 04/2018 – 03/2020
Molecular and Applied Biotechnology, RWTH Aachen University

Bachelor of Science, *German grade: 1.4* | 09/2014 – 03/2018
Molecular and Applied Biotechnology, RWTH Aachen University

ADDITIONAL QUALIFICATIONS

Awards: *Borchers Medal*, RWTH Aachen University, awarded for outstanding doctoral performance
Certificates: Six Sigma Green Belt, GMP
Training: Communication & Team Development, Leadership
Engagement: *TANDEMdok* mentoring programme, *iGEM* research competition

LANGUAGES

English: Fluent (C1)
German: Native

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